

PANCREATIC HORMONES & REGULATION OF BLOOD GLUCOSE

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LEARNING OBJECTIVES

By the end of this lecture, students should be able to:

- ✓ Identify the endocrine part of pancreas & major cell types of islets of Langerhans.
- ✓ Identify the major pancreatic hormones secreted by the islets of Langerhans & describe their functions in maintaining blood glucose levels.
- ✓ Explain the regulation of insulin & glucagon secretion in response to changes in blood glucose levels.
- ✓ Describe the effects of insulin on glucose uptake, glycogenesis & glucose utilization.
- ✓ Describe the effects of glucagon on glycogenolysis & gluconeogenesis.
- ✓ Explain how insulin & glucagon act together to maintain blood glucose homeostasis in the fed & fasting states.
- ✓ Define & classify diabetes mellitus
- ✓ Explain briefly the pathophysiology of type I & type II diabetes.
- ✓ Identify the clinical features, diagnostic tests & complications of diabetes mellitus

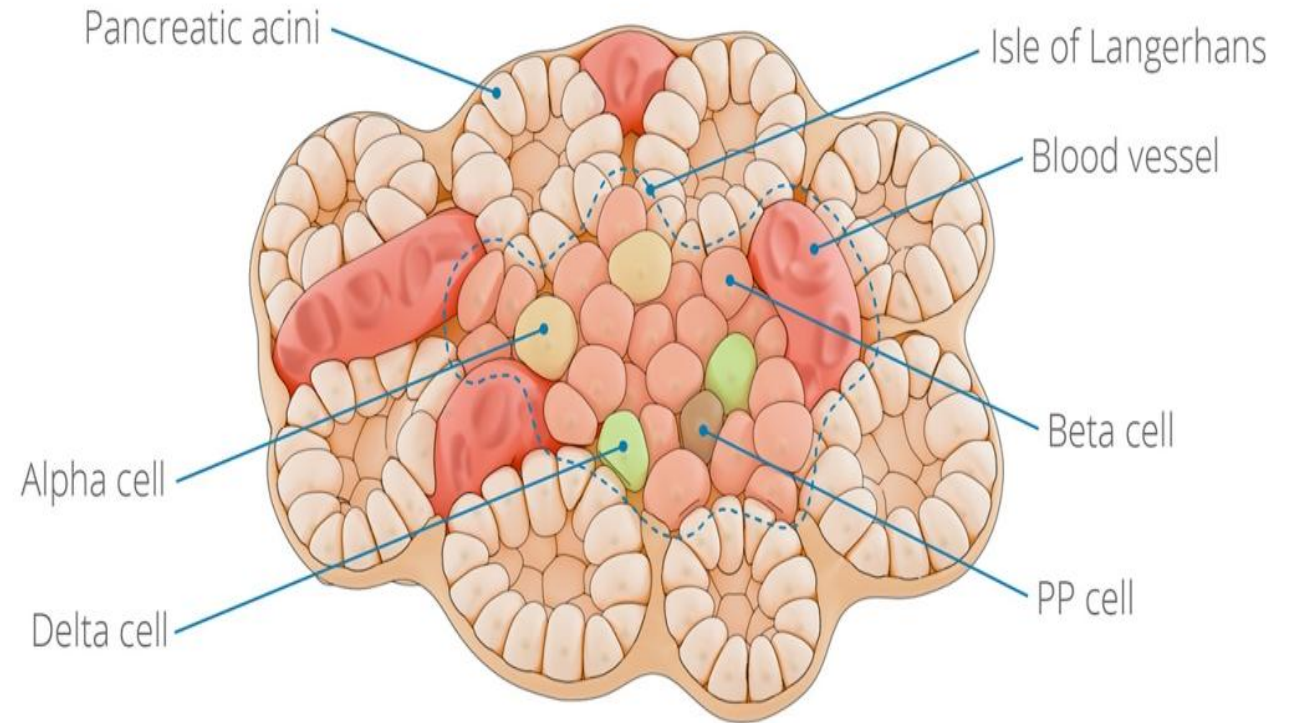
PHYSIOLOGICAL ANATOMY OF PANCREAS

- Pancreas is composed of two major types of tissues i.e.
 - 1) **ACINI** which secrete digestive juices into the duodenum &
 - 2) **ISLETS OF LANGERHANS** which secrete **insulin** & **glucagon** directly into the blood.
- Human pancreas has 1 to 2 million islets of Langerhans (each islet is 0.3mm in diameter) organized around small capillaries into which its cells secrete their hormones.
- The islets contain three major types of cells distinguished from one another by their morphological & staining characteristics i.e.
 - 1) **Alpha cells** which constitute about 25% of total islets cells & secrete **glucagon** that raises blood sugar by signaling liver to release stored glucose.
 - 2) **Beta cells** constitute about 60% of total islets cells & secrete **insulin** & amylin(a hormone often secreted in parallel with insulin) which lowers blood sugar by helping cells to take in glucose from the blood.

Cont..

3) **Delta cells** constituting about 10% of total cells of islets & secrete **somatostatin** which has a balancing effect & inhibits the release of both insulin & glucagon.

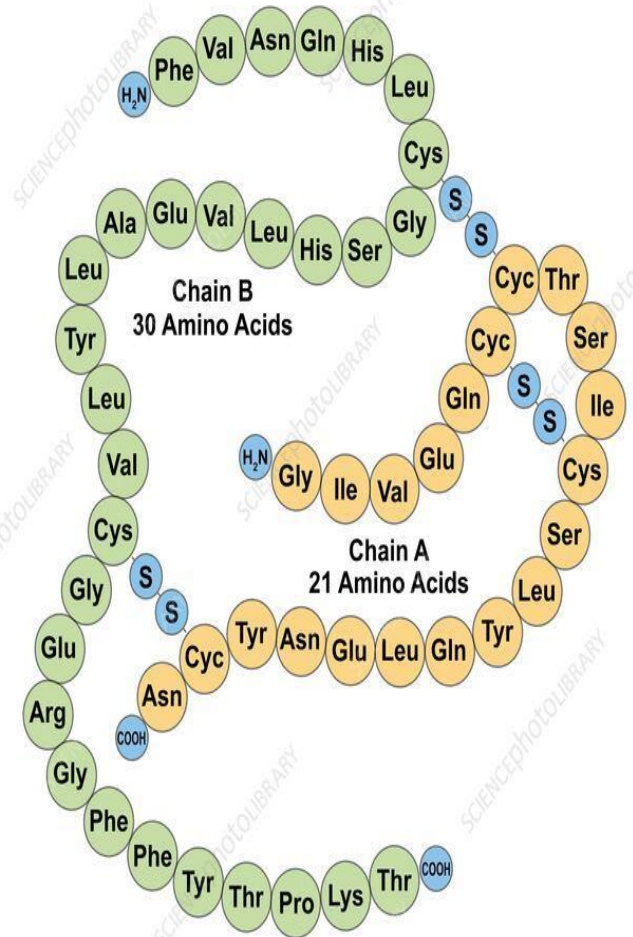
□ In addition to these cells, another type of cells the **PP cells** are also present in small no. & produce hormone called **pancreatic polypeptide** which regulates appetite & gut secretions.



INSULIN

- Secreted by the **beta cells** present in the islets of Langerhans of pancreas.
- It is a polypeptide with 51 amino acids (molecular weight = 5,808) having two amino acids chains i.e. **alpha chain** (contains 21 amino acids) & **beta chain** (contains 30 amino acids).
- Half life of insulin is 5 min whereas basal insulin level in plasma is 10 micro units / milliliter.
- Insulin is synthesized in the rough endoplasmic reticulum of beta cells present in the islets of Langerhans.
- It is synthesized as **preproinsulin** which form **proinsulin**.
- Proinsulin is converted into **insulin & C peptide** (connecting peptide that links alpha & beta chains) through **peptic cleavages**.
- Insulin is degraded in the liver & kidney by the action of enzyme called **insulin protease (insulin degrading enzyme)**.

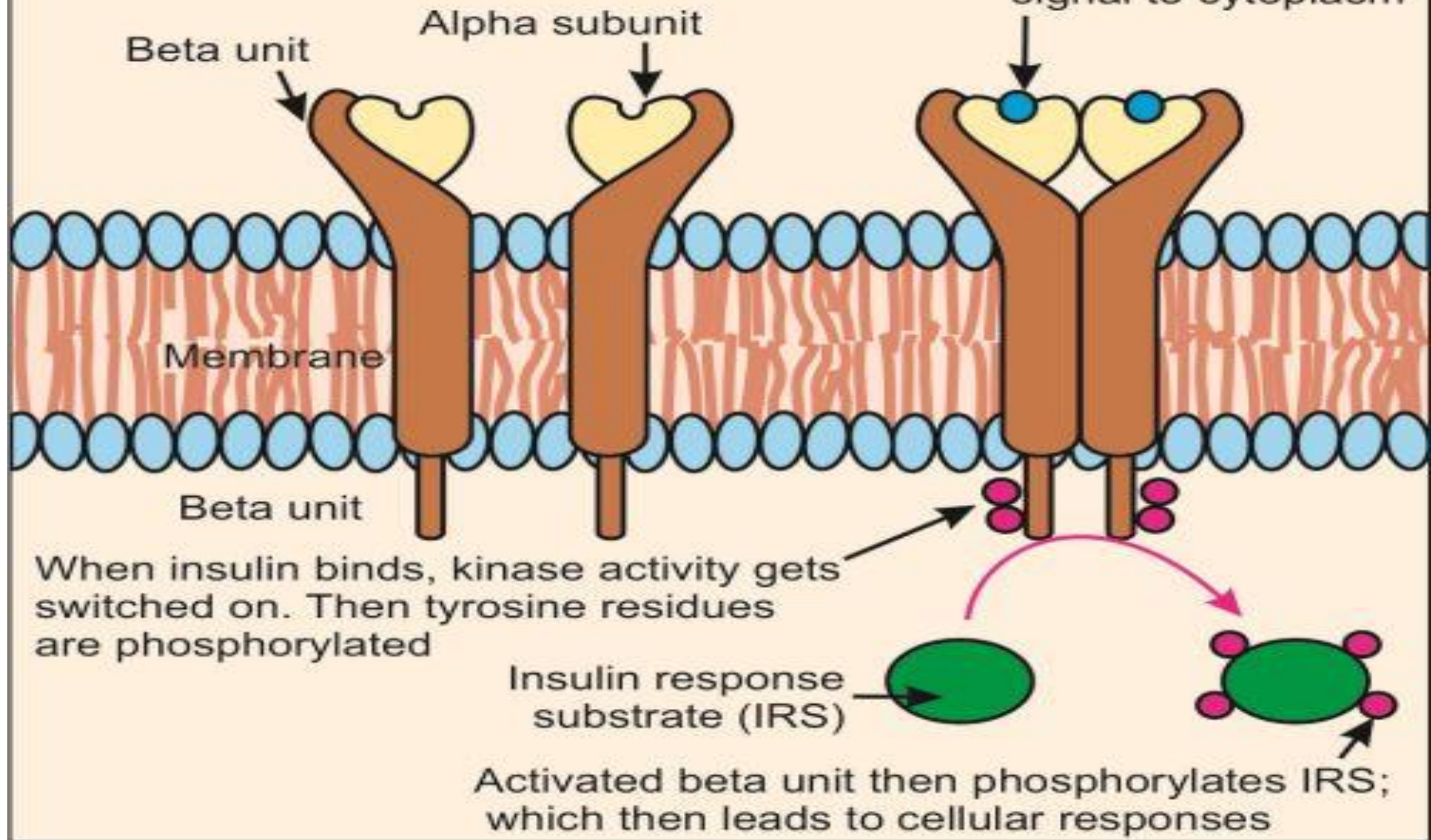
Human Insulin Structure



INSULIN RECEPTOR

- On target cells, insulin binds with insulin receptor (present in almost all the cells of the body) & forms ***insulin-receptor complex***.
- Binding of insulin to insulin receptor is necessary for its removal from circulation & degradation.
- Insulin receptor is a ***glycoprotein*** (molecular weight = 3,40,000) which is a ***tetramer*** of four glycoprotein subunits i.e. ***two alpha subunits*** & ***two beta subunits*** linked together by disulfide bonds.
- Intracellular surfaces of alpha subunits have the enzyme activity i.e. the ***protein kinase (tyrosine kinase)*** activity.
- When insulin binds with alpha subunits of insulin receptor, tyrosine kinase at the beta subunit is activated by means of autophosphorylation.
- Activated tyrosine kinase acts on many intracellular enzymes so that some enzymes are activated while others are inactivated.

When insulin binds with alpha unit, the beta unit transmits signal to cytoplasm



REGULATION OF INSULIN SECRETION

- Insulin secretion is mainly regulated by blood glucose level however, other factors like amino acids, lipid derivatives, gastrointestinal & endocrine hormones & autonomic nervous system can also stimulate insulin secretion.

Effect of blood glucose level on insulin secretion

- When blood glucose level is normal i.e. 80-100 mg/dL, rate of insulin secretion will be low (upto 10 micro units/min).
- However, when blood glucose level increases between 100-120 mg/dL, rate of insulin secretion rises rapidly to 100 micro units/min & when blood glucose level rises above 200 mg/dL, rate of insulin secretion rises very rapidly upto 400 micro units/min.

Biphasic effect of glucose

- Initially when blood glucose level increases after a meal, release of insulin into blood also increases rapidly & within few min conc. of insulin in plasma rises upto 100 microunits/mL from the basal level i.e. 10 microunits/mL.
- Later, within 10 to 15 minutes insulin conc. in blood plasma decreases to half of the value i.e. upto 40 to 50 microunits/mL.
- After 15 to 20 min insulin secretion again rises slowly & steadily reaching maximum between two to two and half hrs due to the formation of new insulin molecules in pancreas.

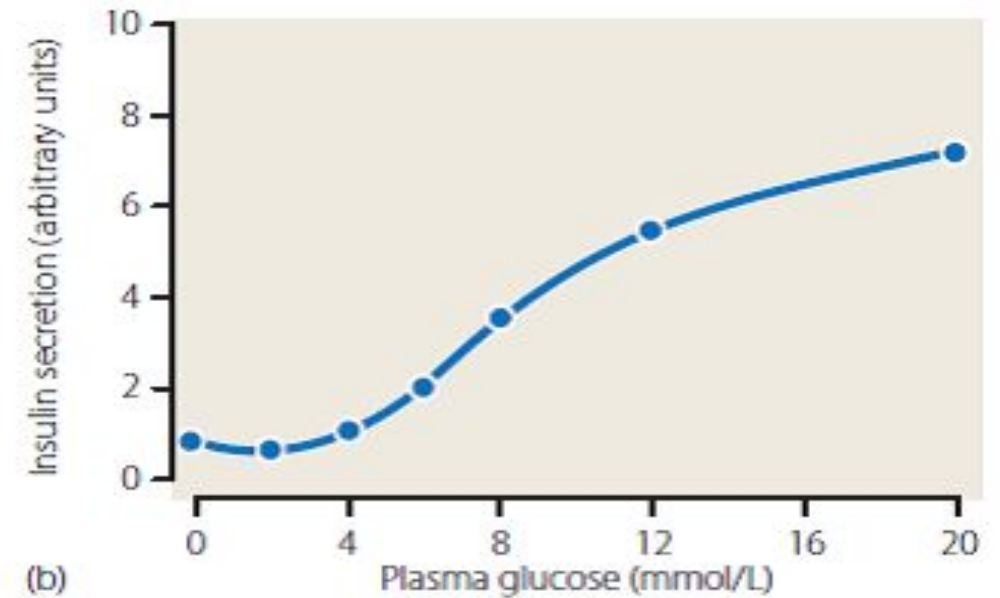
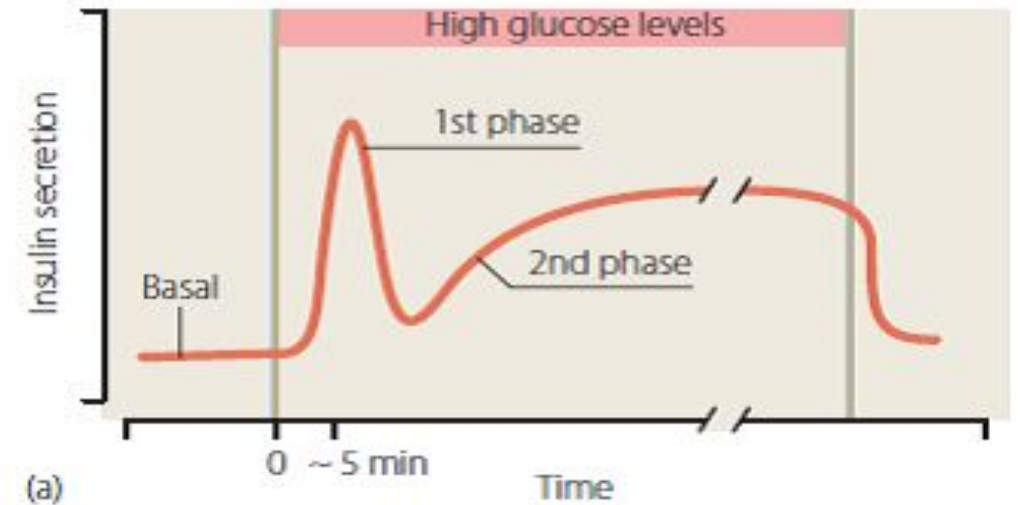


Figure 5.11 (a) The biphasic glucose-stimulated release of insulin from pancreatic islets. (b) The glucose-insulin dose-response curve for islets of Langerhans.

Effect of proteins on insulin secretion

- Amino acids potentiate the action of glucose on insulin secretion however, excess amino acids alone in blood (without any increase in blood glucose level) can slightly increase insulin secretion.

Effect of gastrointestinal hormones & lipid derivatives on insulin secretion

- Some gastrointestinal hormones like **gastrin**, **secretin**, **CCK**, **GIP** & some lipid derivatives like **beta ketoacids** can also stimulate insulin secretion.

Effect of other endocrine hormones on insulin secretion

- Diabetogenic hormones like **glucagon**, **growth hormone** & **cortisol** stimulate insulin secretion indirectly by increasing the blood glucose levels which stimulates beta cells of the islets of Langerhans.
- However, **prolonged hypersecretion** of these hormones causes exhausted beta cells resulting in **diabetes mellitus**.

FUNCTIONS OF INSULIN

- Insulin is an ***anabolic hormone*** involved in ***carbohydrate metabolism*** & also concerned with ***proteins & fats metabolism***.

ROLE OF INSULIN IN CARBOHYDRATE METABOLISM

- Insulin is the only ***antidiabetic hormone*** (reduces blood glucose level) secreted in the body.
- Effects of insulin on carbohydrate metabolism are:

1. Increase transport & uptake of glucose by the cells

- Insulin facilitates transport of glucose into the cells by increasing permeability of cell membrane to glucose via ***GLUT4 translocation***.
- It also increases the no. of ***glucose transporters*** particularly GLUT4 in the cell membrane.

2. Promote peripheral utilization of glucose

- Insulin also promotes the peripheral utilization of glucose i.e. the glucose which enters the cell is oxidized immediately in the presence of insulin.

3. Promote glycogenesis & inhibits glycogenolysis

- Insulin promotes conversion of glucose into glycogen by activating enzymes necessary for glycogenesis which is then stored in the muscles & liver.
- In liver when glycogen content increases beyond its storing capacity, insulin causes conversion of glucose into fatty acids.
- Insulin also inhibits the breakdown of glycogen into glucose in the muscles & liver.

4. Inhibit gluconeogenesis

- Insulin also inhibits the formation of glucose from proteins by inhibiting the release of amino acids from muscles & also inhibits the activities of enzymes necessary for gluconeogenesis.

GLUCOSE TRANSPORTERS

- Usually, glucose is transported into the cells via ***sodium-glucose symport pump*** however, most of the cells have another type of transport proteins named as ***glucose transporters (GLUT)***.
- Among GLUT1-7, GLUT4 is most important due to its insulin sensitivity & speedy transport of glucose.
- It is present in cytoplasmic vesicles & in large no. in muscle fibers & adipose cells.
- When ***insulin-receptor complex*** is formed in the cell membrane of these cells, cytoplasmic vesicles containing GLUT4 are attracted & released in the membrane.
- GLUT4 then starts transporting glucose molecules from extracellular fluid into the cell.

ROLE OF INSULIN IN PROTEIN METABOLISM

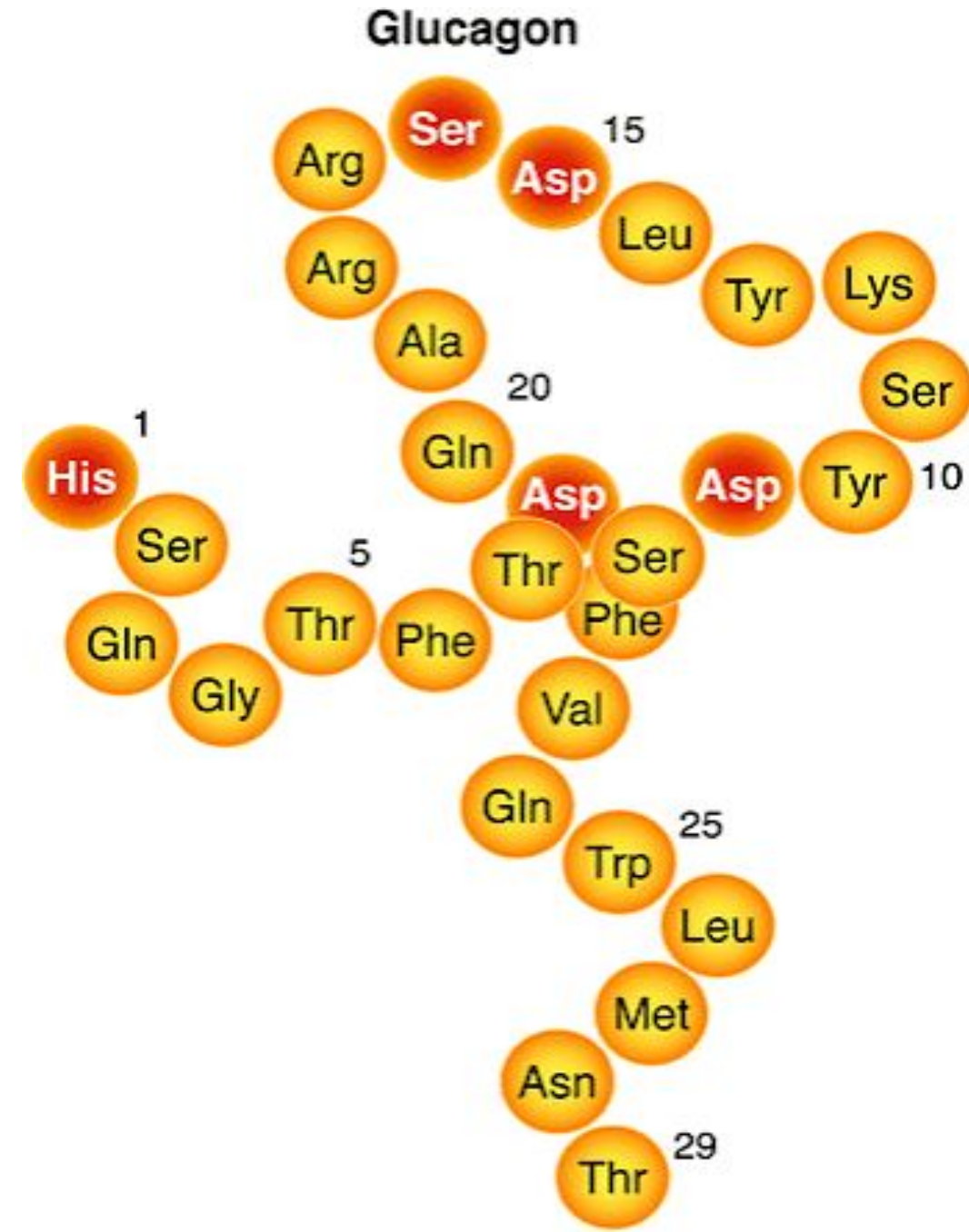
- Insulin facilitates transport of amino acids into the cell from the blood by increasing permeability of cell membrane for amino acids.
- It also accelerates protein synthesis by influencing transcription of DNA & translation of mRNA & also inhibits degradation of proteins by inhibiting the activity of enzymes involved in protein catabolism.
- Insulin prevents conversion of proteins into glucose.

ROLE OF INSULIN IN FAT METABOLISM

- Insulin stimulates synthesis of fats by promoting transport of excess glucose into the cells especially the liver cells which is utilized for the production of fatty acids & triglycerides.
- It also stimulates enzymes necessary for the conversion of glucose into fatty acids & fatty acids into triglycerides.
- Insulin facilitates transport of fatty acids into adipose tissue & promotes storage of fats by inhibiting enzymes involved in degradation of triglycerides.

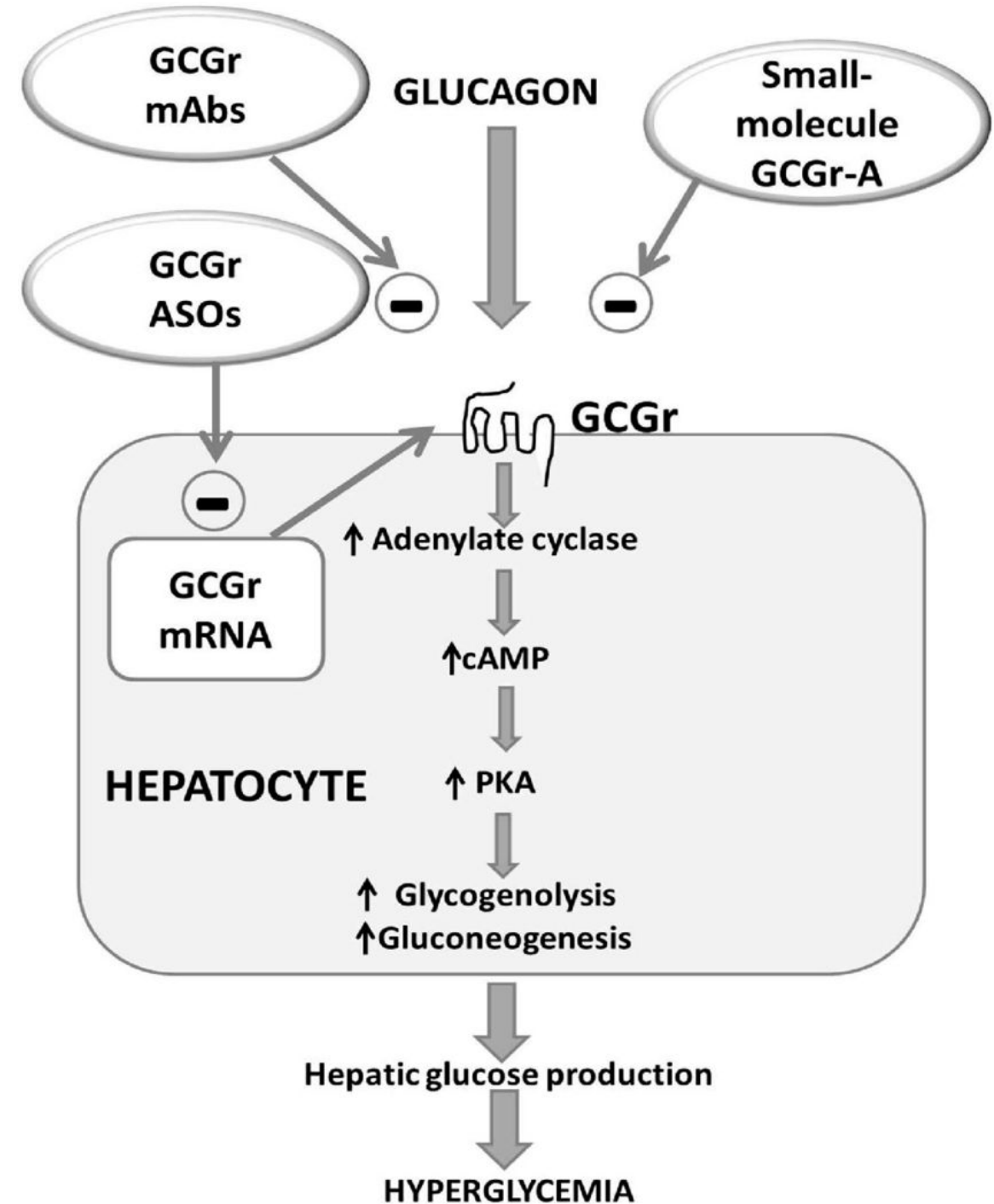
GLUCAGON

- It is secreted by the **alpha cells** present in the islets of Langerhans in pancreas & also from **A cells of stomach & L cells of intestine**.
- It is a **polypeptide of 29 amino acids** (molecular weight = 3,485) & has a half life of 3 to 6 min.
- It is synthesized from **preproglucagon** in the alpha cells of islets of Langerhans which is converted into **proglucagon** & proglucagon is finally converted into glucagon.
- Circulating glucagon in the blood is degraded by actions of enzymes such as **serine** & **cysteine proteases**.



GLUCAGON RECEPTOR

- Glucagon receptor is a peptide whose molecular weight is 62,000.
- On target cells especially the liver cells, glucagon combines with its ***G protein coupled receptor*** & activates ***adenyl cyclase*** which forms ***cyclic adenosine monophosphate (cAMP)***.
- cAMP then brings out the actions of glucagon by activating a cascade of chemical reactions that breakdown glycogen into free glucose.



REGULATION OF GLUCAGON SECRETION

- Secretion of glucagon is controlled mainly by glucose & amino acids levels in the blood.

Effect of blood glucose level on glycogen secretion

- When blood glucose level decreases below 80mg/dL, alpha cells of islets are activated which secrete more glucagon.
- When blood glucose level increases, alpha cells are inhibited which decreases glucagon secretion.

Effect of amino acids level on glucagon secretion

- Increased amino acids level in the blood stimulates glucagon secretion which then converts amino acids into glucose & blood glucose level is increased.

Effect of other factors on glucagon secretion

- Factors which stimulate glucagon secretions are exercise, stress & enzymes like gastrin, cholecystokinin & cortisol.
- Enzymes (like insulin, somatostatin), free fatty acids & ketones are the factors that inhibit glucagon secretion.

FUNCTIONS OF GLUCAGON

- Action of glucagon is antagonistic to the action of insulin i.e. glucagon increases blood glucose level, peripheral utilization of lipids & conversion of proteins into glucose.

ROLE OF GLUCAGON IN CARBOHYDRATE METABOLISM

- It increases blood glucose level by increasing glycogenolysis into glucose in liver(not in muscles).

ROLE OF GLUCAGON IN PROTEIN METABOLISM

- It also increases gluconeogenesis in liver by increasing transport of amino acids into liver cells which are then utilized for glucose production.

ROLE OF GLUCAGON IN FAT METABOLISM

- Glucagon increases lipolysis by increasing release of free fatty acids from adipose tissue for peripheral utilization.
- This lipolytic activity of glucagon causes formation of ketone bodies in the liver.

Insulin vs Glucagon

● Beta cells of pancreas

● Blood glucose high
(after eating)

● Decreases blood sugar

● Moves glucose into body
cells

● Stores glucose as
glycogen (glycogenesis)

● Promotes fat storage

● Anabolic (building &
storing)

● Prevents hyperglycemia

● Storage hormone

● Fed state hormone

● Alpha cells of pancreas

● Blood glucose low
(fasting/exercise)

● Increases blood sugar

● Releases glucose from
liver to blood

● Breaks glycogen into
glucose (glycogenolysis)

● Promotes fat breakdown

● Catabolic (breaking &
releasing)

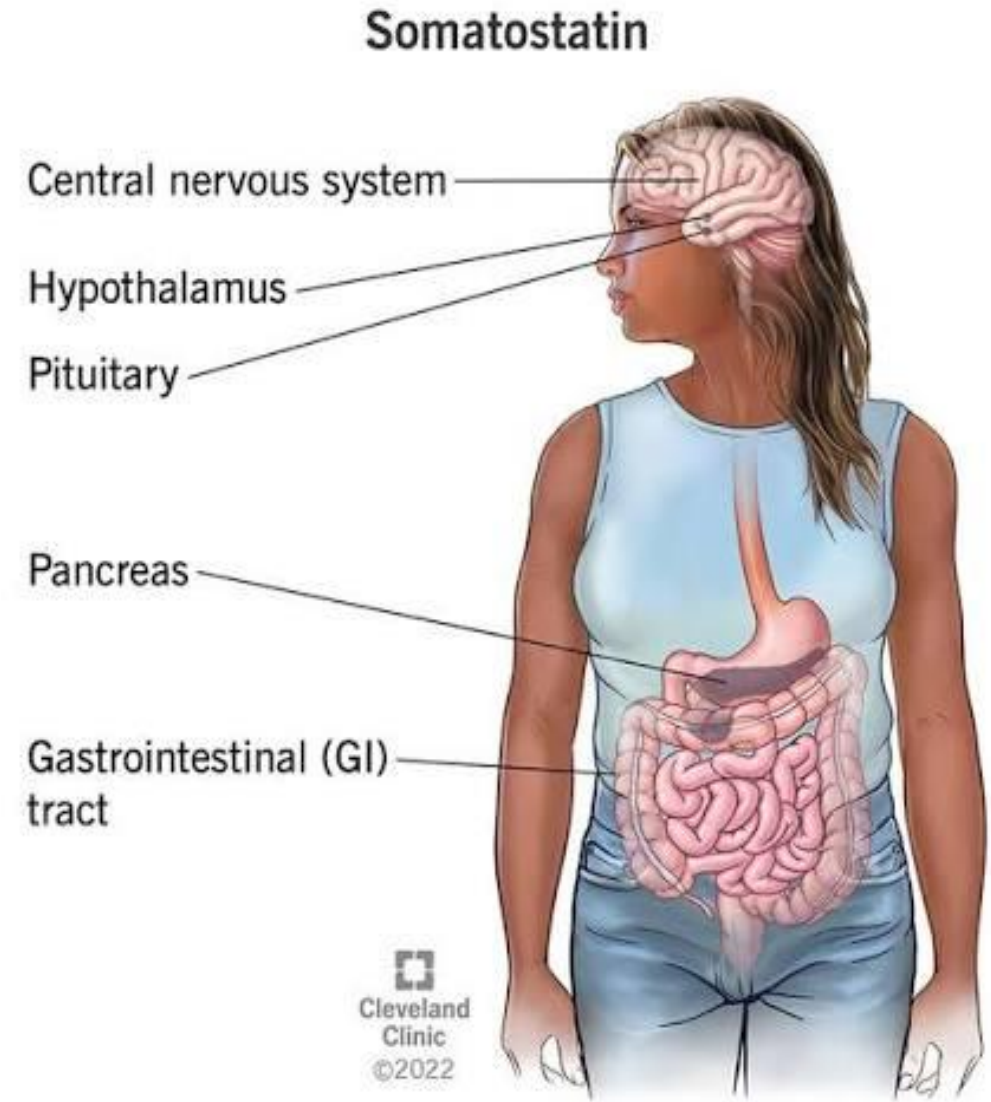
● Prevents hypoglycemia

● Release hormone

● Fasting state hormone

SOMATOSTATIN

- It is secreted from hypothalamus, delta cells in the islets of Langerhans & also from D cells in stomach & upper part of small intestine.
- It is a **polypeptide** synthesized in two forms i.e. **somatostatin-14** (chain of 14 amino acids) & **somatostatin-28** (having 28 amino acids).
- Somatostatin is formed from **prosomatostatin** which is converted mainly into **somatostatin-14 in the islets of Langerhans** & **somatostatin-28 in the intestine**.



REGULATION OF SOMATOSTATIN

- Secretion of pancreatic somatostatin is stimulated by glucose, amino acids & cholecystokinin(CCK).
- Somatostatin secretion in the stomach & small intestine is activated by the presence of chyme containing glucose & proteins.

FUNCTIONS OF SOMATOSTATIN

- Somatostatin ***inhibits alpha & beta cells*** of islets of langerhan's resulting in reduced secretion of both glucagon & insulin.
- It also decreases secretion of gastrointestinal hormones like ***gastrin, cholecystokinin*** (CCK), ***gastric inhibitory polypeptide*** (GIP) & ***vasoactive intestinal peptide*** (VIP).
- Somatostatin produced by hypothalamus inhibits the secretion of growth hormone(GH) & thyroid stimulating hormone (TSH) from the pituitary gland.

PANCREATIC POLYPEPTIDE

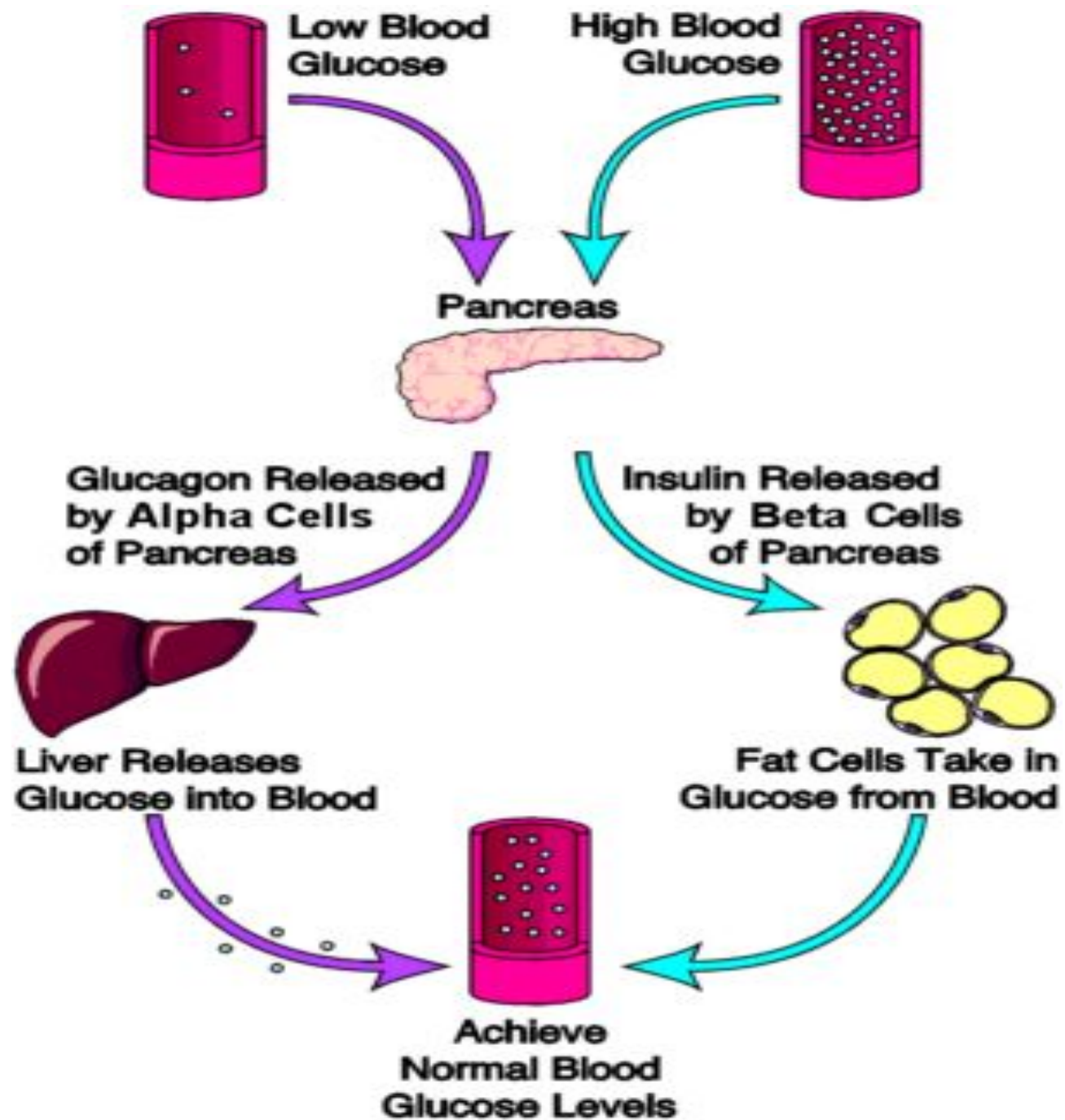
- Pancreatic polypeptide is secreted by PP cells present in the islets of Langerhans of pancreas.
- It is a polypeptide with 36 amino acids & also present in small intestine.
- Pancreatic polypeptide has a half life of 5 min.
- It is synthesized from prepropancreatic polypeptide in the PP cells of islets of Langerhans.

REGULATION OF SECRETION & FUNCTION OF PANCREATIC POLYPEPTIDE

- Secretion of pancreatic polypeptide is stimulated by the presence of more protein content in the chyme of small intestine.
- Its exact function is unclear however, it is believed that it stimulates the secretion of glucagon from alpha cells of islets of Langerhans.

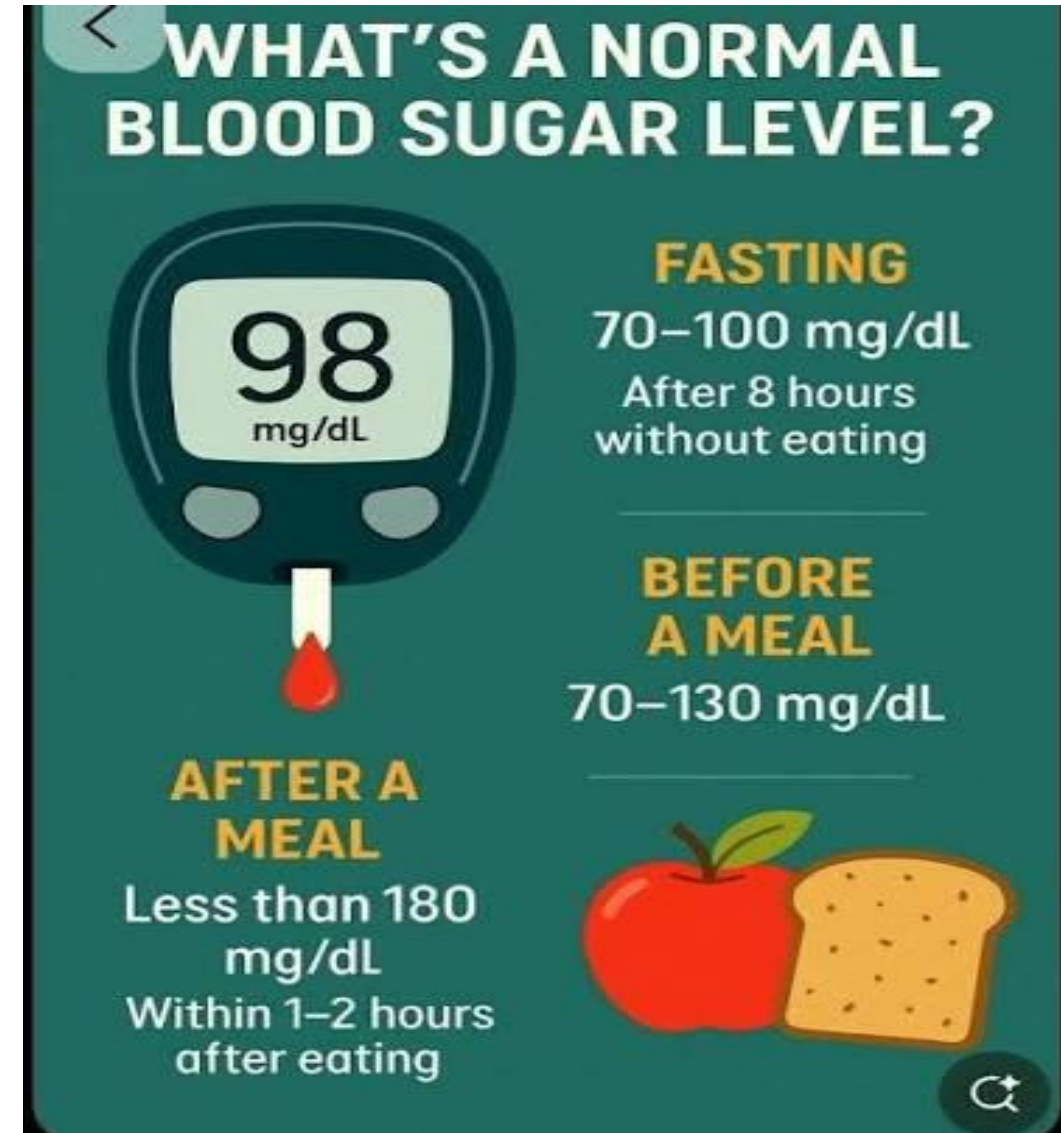
REGULATION OF BLOOD GLUCOSE LEVEL

- Regulation of blood glucose level is very important as glucose is the only nutrient which provides energy to many tissues of the body including brain tissues, retina & germinal epithelium of gonads etc.
- Blood glucose regulation is controlled through liver & muscles under the influence of pancreatic hormones i.e. insulin & glucagon.
- Many other hormones are also involved in the regulation of blood sugar level however, ***insulin*** is the only hormone that decreases blood sugar level so it is termed as ***anti diabetogenic hormone***.
- Hormones that increase the blood sugar level are termed as ***diabetogenic*** or ***anti-insulin hormones***.



NORMAL BLOOD GLUCOSE LEVEL

- ❑ In the morning after overnight fasting, blood glucose level is low ranging between 70 to 110mg/dL.
- ❑ Between first & second hour after taking meal(postprandial), blood sugar level rises to 140mg/dL from an average of 100mg/dL.
- ❑ However, blood sugar level is returned back to normal at the end of second hour after meals.



Blood Glucose Chart

Mg/DL

Fasting

After Eating

**2-3 Hours
After Eating**

Normal

80-100

170-200

120-140

**Impaired
Glucose**

101-125

190-230

140-160

Diabetic

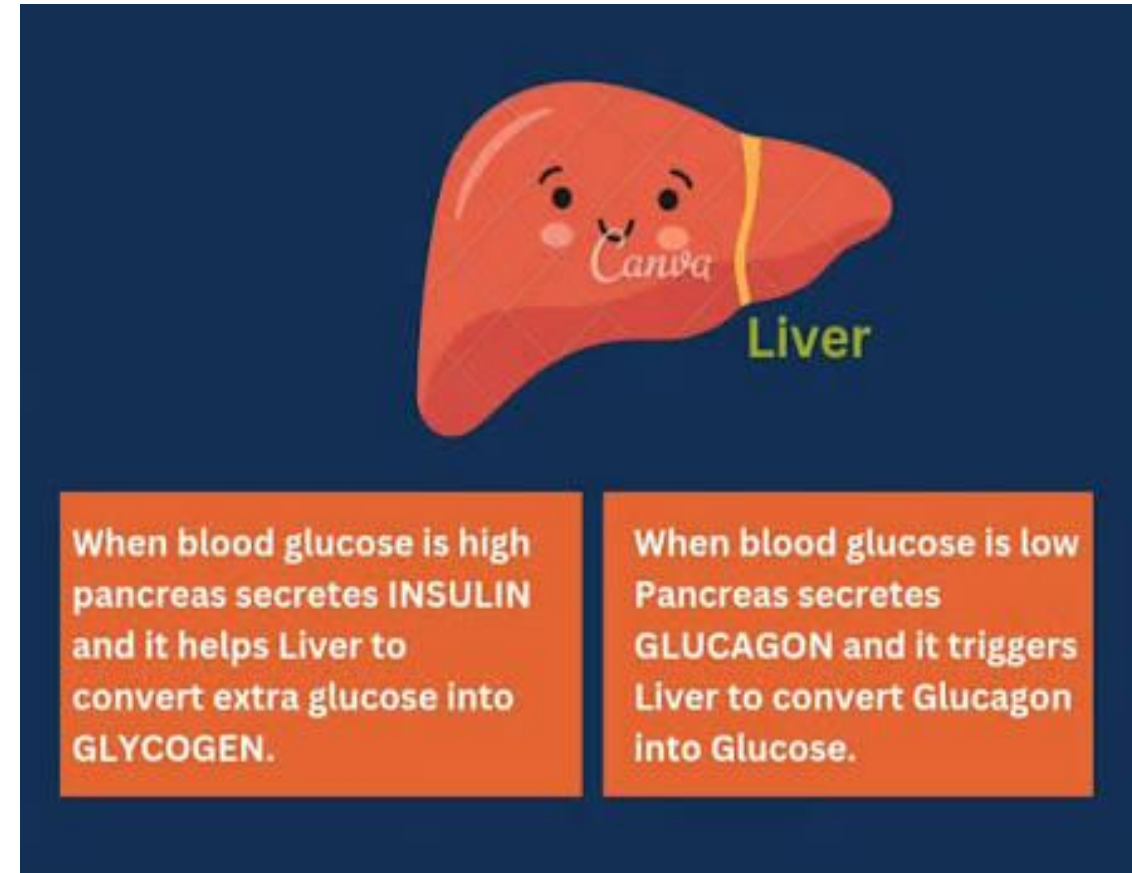
126+

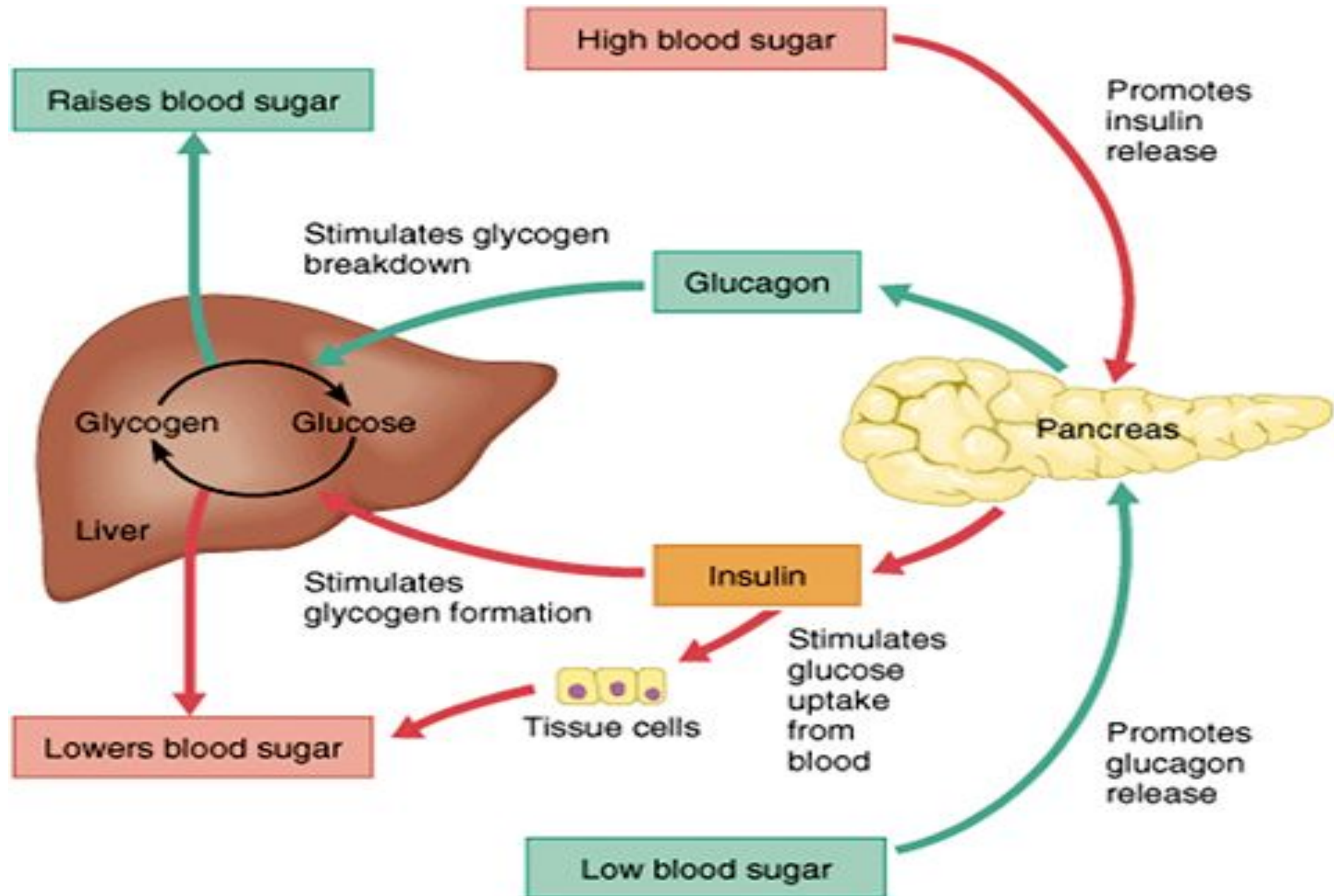
220-300

200+

ROLE OF LIVER IN THE MAINTENANCE OF BLOOD GLUCOSE LEVEL

- ❑ Liver acts as an important *glucose buffer system* i.e. when blood glucose level increases after a meal, the excess glucose is converted into glycogen to be stored in the liver.
- ❑ Afterwards, when blood glucose level falls, the stored glycogen in the liver is converted into glucose & released in the blood for peripheral utilization.
- ❑ This storage of glycogen & release of glucose from the liver is regulated mainly by two main pancreatic hormones i.e. insulin & glucagon.





ROLE OF OTHER HORMONES IN MAINTAINING BLOOD GLUCOSE LEVEL

□ Other hormones which increases the blood glucose level are:

1) Growth hormone

- It is secreted by the pituitary gland & prevent blood sugar from falling too low during fasting or exercise.
- It reduces the glucose uptake in muscles & stimulates the breakdown of stored fats for energy sparing glucose for the nervous system.

2) Thyroxine

- It is produced by the thyroid gland & enhances glycogen breakdown to keep baseline energy & glucose available.
- It also increases the rate of carbohydrate absorption from the digestive tract.

3) Cortisol

- It is released by the adrenal glands during stress & stimulates gluconeogenesis in the liver using amino-acids & fats.
- It also makes muscles resistant to insulin so that more glucose remain in the bloodstream for the brain.

4) Adrenaline

- Released during fight or flight situations & stimulates glycogenolysis in the liver to release glucose into the blood stream.

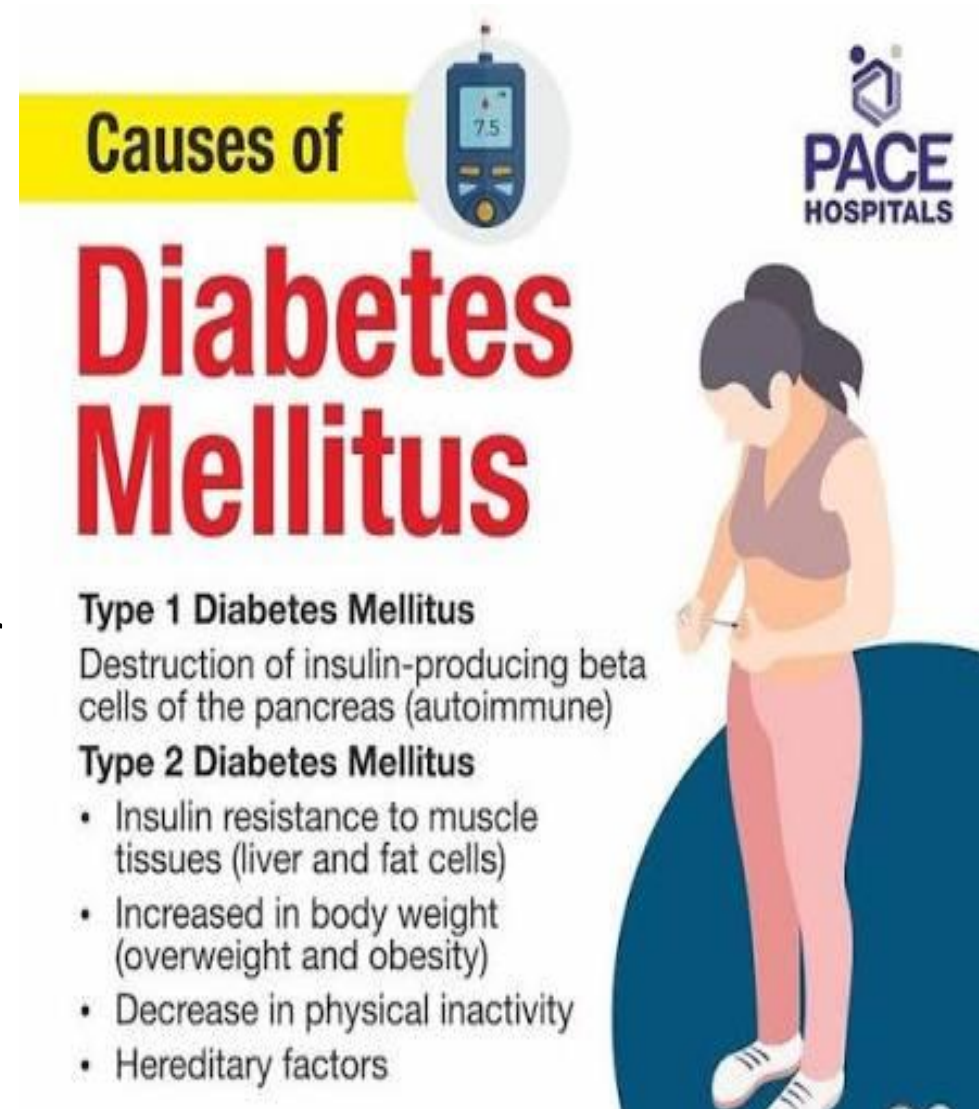
APPLIED PHYSIOLOGY

DIABETES MELLITUS

- It is a metabolic disorder characterized by high blood glucose level develops mostly due to deficiency of insulin.

CLASSIFICATION OF DIABETES MELLITUS

- Diabetes can be classified as primary diabetes or secondary diabetes.
- Primary diabetes is unrelated to another disease whereas, secondary diabetes results due to damage or disease of pancreas by another disease or factor.
- Primary diabetes can further be classified into two types:
 1. Type I diabetes mellitus
 2. Type II diabetes mellitus



TYPE I DIABETES MELLITUS (INSULIN DEPENDENT DIABETES MELLITUS)

- It is caused due to ***deficiency of insulin*** because of ***destruction of beta cells*** in the islets of Langerhans.
- Type I diabetes may occur at any age but usually occurs before 40 years of age & the affected person require insulin injection.
- Type I diabetes develops rapidly & progresses at a rapid rate, if it develops at infancy or childhood it is termed as ***juvenile diabetes***.
- It can be caused due to ***degeneration or destruction of beta cells*** by viral infection, ***congenital disorder of beta cells*** or destruction of beta cells by autoimmune diseases or due to the development of antibodies against beta cells.

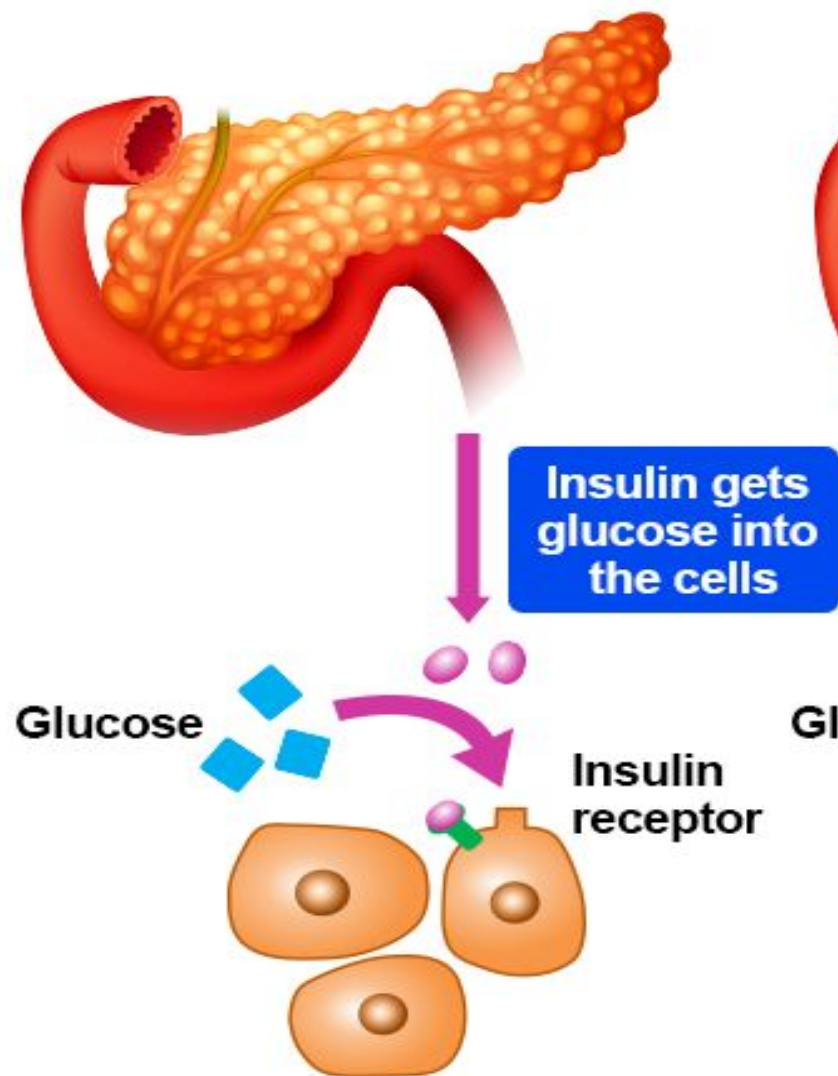
TYPE II DIABETES MELLITUS

(NON-INSULIN-DEPENDENT DIABETES MELLITUS)

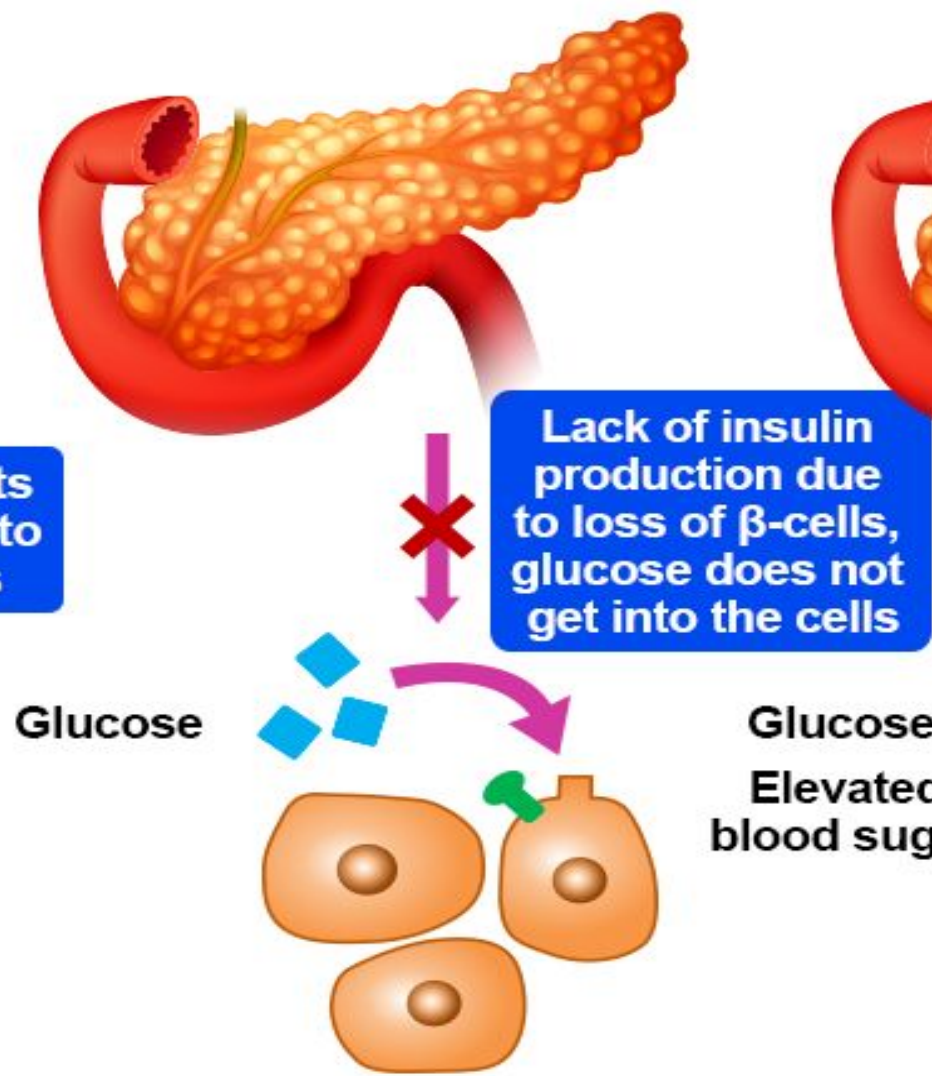
- Type II diabetes is caused due to ***insulin resistance*** i.e. due to failure of insulin receptors to respond to insulin so the body is unable to use insulin.
- In this type of diabetes mellitus, structure & function of beta cells & blood insulin level are normal but insulin receptors may be less, absent or abnormal resulting in insulin resistance.
- About 90% of diabetic patients have this type of diabetes & usually occurs after 40 years.
- In most cases, type II diabetes can be controlled by oral hypoglycemic drugs & only some forms require insulin.
- Type II diabetes may or may not be associated with ketosis but usually associated with obesity.
- Common causes of insulin resistance include genetic disorders, lifestyle changes like bad eating habits, less physical activity leading to obesity & stress.

Diabetes Mellitus

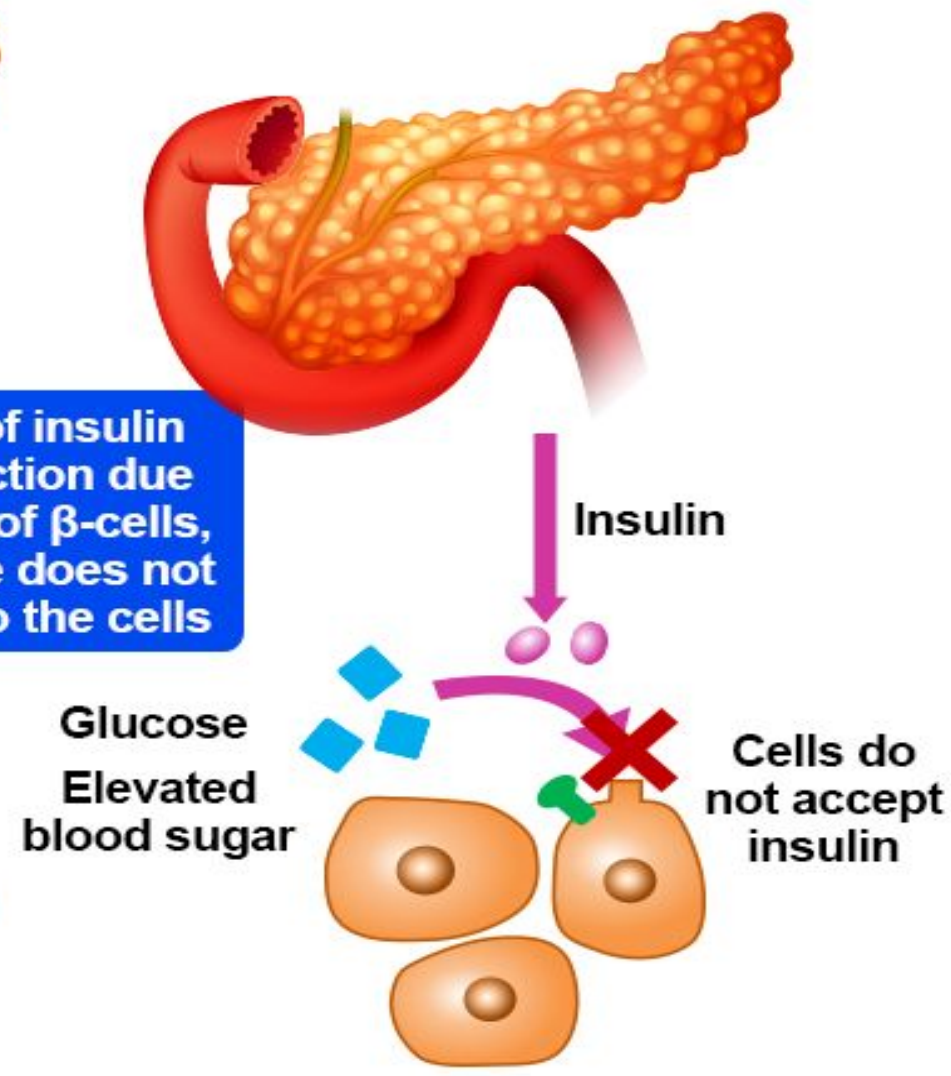
NORMAL



TYPE 1 DIABETES



TYPE 2 DIABETES



Gestational diabetes

- This type of diabetes occurs during pregnancy & is due to many factors such as hormones secreted during pregnancy, obesity & lifestyle before & during pregnancy.
- Gestational diabetes usually disappears after delivery of baby but these women are at high risk of developing type II diabetes later in life.

Prediabetes (border-line diabetes)

- It is the stage characterized by blood glucose levels between normal & diabetic conditions.
- In prediabetes, the person does not show observable symptoms of diabetes, but there is an increase in blood glucose level.
- Although prediabetes is reversible, but the prediabetic people are at high risk to develop type II diabetes mellitus.

SECONDARY DIABETES MELLITUS

- It is a rare type of diabetes mellitus & only about 2% of total diabetic patients have secondary diabetes.
- It may be temporary or permanent due to the underlying cause.

Causes:

- Endocrine disorders like gigantism, acromegaly & Cushing's syndrome cause excess stimulation of beta islets cells due to hyperglycemia resulting in exhaustion & degeneration of these cells leading to permanent diabetes mellitus.
- Surgical removal of pancreas(pancreatectomy) or Pancreatic damage due to pancreatitis, cystic fibrosis & hemochromatosis.
- Liver diseases such as fatty liver or hepatitis C & autoimmune diseases.
- Excessive use of drugs like antihypertensive drugs, oral contraceptives, chemotherapy etc & excessive intake of alcohol.

SIGNS & SYMPTOMS OF DIABETES MELLITUS

1. Glucosuria (glucose in urine)
2. Osmotic diuresis leading to polyuria
3. Polydipsia (excessive thirst)
4. Polyphagia (excessive hunger)
5. Asthenia (less physical strength & energy or body weakness)
6. Acidosis (ketoacidosis)
7. Acetone breathing (fruity or acetone smelling breath)
8. Kussmaul breathing (deep, rapid breathing pattern)
9. Circulatory shock
10. Coma

DIAGNOSTIC TESTS FOR DIABETES

- ☐ Fasting blood glucose
- ☐ Glucose tolerance test
- ☐ Glycosylated hemoglobin

Glycosylated hemoglobin test is usually done to monitor the glycemic control of patients who have already been diagnosed with diabetes mellitus.



COMPLICATIONS OF DIABETES MELLITUS

- Complications of diabetes mellitus develop due to dysfunction & injury of many tissues due to prolonged hyperglycemia.
- These complications are directly proportional to the degree & duration of hyperglycemia.
- In patients with well controlled diabetes, the onset & rate of progression of these complications can be delayed or reduced.
- Common complications of diabetes mellitus are:
 1. Cardiovascular complications like hypertension, myocardial infarction
 2. Degenerative changes in retina(diabetic retinopathy)
 3. Degenerative changes in kidney(diabetic nephropathy)
 4. Degeneration of autonomic & peripheral nerves(diabetic neuropathy)

TREATMENT FOR DIABETES MELLITUS

TYPE I DIABETES MELLITUS

- Subcutaneous injection of ***exogenous insulin***.

TYPE II DIABETES MELLITUS

- Type II diabetes is treated with oral ***hypoglycemic drugs***.
- Patients with long lasting severe diabetes may require a combination of oral hypoglycemic drugs with insulin to control hyperglycemia.

Management and Treatment

Lifestyle Changes



Healthy diet



Regular exercise



Weight management

Monitoring



Regularly checking blood sugar levels is crucial to managing diabetes

Medications



Type 1 Diabetes: Insulin therapy is necessary



Type 2 Diabetes: Oral medications, non-insulin injectables, or insulin therapy may be prescribed

HYPERACTIVITY OF ISLETS CELLS

1. Hyperactivity of beta cells (hyperinsulinism)

- Hyperinsulinism is the hypersecretion of insulin resulting in hypoglycemia(blood glucose level falls below 50mg/dL) & occurs due to the tumor of beta cells in the islets of Langerhans.

2. Hyperactivity of delta cells

- Tumor of D cells in the islets of Langerhans causes hypersecretion of somatostatin resulting in hyperglycemia & other symptoms of diabetes mellitus.

3. Hyperactivity of alpha cells (hyperglucagonemia)

- Hyperactivity of pancreatic alpha islet cells due to tumors leads to over-production & excessive secretion of glucagon(hyperglucagonemia) which causes raised blood glucose(hyperglycemia) levels by stimulating hepatic glucose production.

POST ASSESSMENT

Q1. Which pancreatic cells primarily secrete insulin?

- a) Alpha cells
- b) Beta cells
- c) Delta cells
- d) Pp cells

Q2. Which hormone is most important in preventing blood glucose from falling excessively during fasting?

- a) Insulin
- b) Glucagon
- c) Somatostatin
- d) Pancreatic polypeptide

Cont..

Q3. After a carbohydrate rich meal, which hormonal change is expected?

- a) Increased glucagon & decreased insulin
- b) Increased insulin & increased glucagon
- c) Decreased insulin & decreased glucagon
- d) Increased insulin & decreased glucagon.

Q4. A patient with diabetes has HbA1c level of 10%. What does HbA1c primarily indicate?

- a) Blood glucose level at the time of testing
- b) Amount of glucagon in the blood
- c) Average blood glucose over the last several weeks/months
- d) Insulin concentration in the blood after a meal

— THANK YOU —

..... FROM THE

BOTTOM

OF MY

PANCREAS

..... WAIT, THAT'S NOT RIGHT